Core Hours

Knowledge Unit	CS Core	KA Core
Fundamentals	2.5 + 0.25 (SEP) + 0.25 (SF)	
Networked Applications	3.5 + 0.25 (SEP) + 0.25 (PDC)	
Reliability Support		5.75 + 0.25 (SF)
Routing And Forwarding		4
Single-Hop Communication		3
Mobility Support		4
Network Security		2.25 + 0.5 (SEC) + 0.25 (SEP)
Emerging Topics		4
Total	**7**	**24**

The shared hours correspond to overlapping concepts that are covered from a networking perspective and are only counted here.

Knowledge Units

NC-Fundamentals: Fundamentals

CS Core:

1. Importance of networking in contemporary computing, and associated challenges. (See also: SEP-Context, SEP-Privacy)
2. Organization of the internet (e.g., users, Internet Service Providers, autonomous systems, content providers, content delivery networks)
3. Switching techniques (e.g., circuit and packet)
4. Layers and their roles (application, transport, network, datalink, and physical)
5. Layering principles (e.g., encapsulation and hourglass model) (See also: SF-Foundations)
6. Network elements (e.g., routers, switches, hubs, access points, and hosts)
7. Basic queueing concepts (e.g., relationship with latency, congestion, service levels, etc.)

Illustrative Learning Outcomes:

CS Core:

1. Articulate the organization of the internet.
2. List and define the appropriate network terminology.

3. Describe the layered structure of a typical networked architecture.
4. Identify the different types of complexity in a network (edges, core, etc.).

NC-Applications: Networked Applications

CS Core:

1. Naming and address schemes (e.g., DNS, and Uniform Resource Identifiers)
2. Distributed application paradigms (e.g., client/server, peer-to-peer, cloud, edge, and fog) (See also: PDC-Communication, PDC-Coordination)
3. Diversity of networked application demands (e.g., latency, bandwidth, and loss tolerance) (See also: PDC-Communication, SEP-Sustainability, SEP-Context)
4. Coverage of application-layer protocols (e.g., HTTP)
5. Interactions with TCP, UDP, and Socket APIs (See also: PDC-Programs)

Illustrative Learning Outcomes:

CS Core:

1. Define the principles of naming, addressing, resource location.
2. Analyze the needs of specific networked application demands.
3. Describe the details of one application layer protocol.
4. Implement a simple client-server socket-based application.

NC-Reliability: Reliability Support

KA Core:

1. Unreliable delivery (e.g., UDP)
2. Principles of reliability (e.g., delivery without loss, duplication, or out of order) (See also: SF-Reliability)
3. Error control (e.g., retransmission, error correction)
4. Flow control (e.g., stop and wait, window based)
5. Congestion control (e.g., implicit and explicit congestion notification)
6. TCP and performance issues (e.g., Tahoe, Reno, Vegas, Cubic)

Illustrative Learning Outcomes:

KA Core:

1. Describe the operation of reliable delivery protocols.
2. List the factors that affect the performance of reliable delivery protocols.
3. Describe some TCP reliability design issues.
4. Design and implement a simple reliable protocol.

NC-Routing: Routing and Forwarding

KA Core:

1. Routing paradigms and hierarchy (e.g., intra/inter domain, centralized and decentralized, source routing, virtual circuits, QoS)
2. Forwarding methods (e.g., forwarding tables and matching algorithms)
3. IP and Scalability issues (e.g., NAT, CIDR, BGP, different versions of IP)

Illustrative Learning Outcomes:

KA Core:

1. Describe various routing paradigms and hierarchies.
2. Describe how packets are forwarded in an IP network.
3. Describe how the Internet tackles scalability challenges. .

NC-SingleHop: Single Hop Communication

KA Core:

1. Introduction to modulation, bandwidth, and communication media
2. Encoding and Framing
3. Medium Access Control (MAC) (e.g., random access and scheduled access)
4. Ethernet and WiFi
5. Switching (e.g., spanning trees, VLANS).
6. Local Area Network Topologies (e.g., data center, campus networks).

Illustrative Learning Outcomes:

KA Core:

1. Describe some basic aspects of modulation, bandwidth, and communication media.
2. Describe in detail a MAC protocol.
3. Demonstrate understanding of encoding and framing solution tradeoffs.
4. Describe details of the implementation of Ethernet.
5. Describe how switching works.
6. Describe one kind of a LAN topology.

NC-Security: Network Security

KA Core:

1. General intro about security (Threats, vulnerabilities, and countermeasures) (See also: SEP-Security, SEC-Foundations, SEC-Engineering)
2. Network specific threats and attack types (e.g., denial of service, spoofing, sniffing and traffic redirection, attacker-in-the-middle, message integrity attacks, routing attacks, ransomware, and traffic analysis) (See also: SEC-Foundations, SEC-Engineering)
3. Countermeasures (: SEC-Foundations, SEC-Crypto, SEC-Engineering)
    a. Cryptography (e.g. SSL, TLS, symmetric/asymmetric)
    b. Architectures for secure networks (e.g., secure channels, secure routing protocols, secure DNS, VPNs, DMZ, Zero Trust Network Access, hyper network security, anonymous communication protocols, isolation)
    c. Network monitoring, intrusion detection, firewalls, spoofing and DoS protection, honeypots, tracebacks, BGP Sec, RPKI

Illustrative Learning Outcomes:

KA Core:

1. Describe some of the threat models of network security.

2. Describe specific network-based countermeasures.
3. Analyze various aspects of network security from a case study.

NC-Mobility: Mobility

KA Core:

1. Principles of cellular communication (e.g., 4G, 5G)
2. Principles of Wireless LANs (mainly 802.11)
3. Device to device communication (e.g., IoT communication)
4. Multi-hop wireless networks (e.g., ad hoc networks, opportunistic, delay tolerant)

Illustrative Learning Outcomes:

KA Core:

1. Describe some aspects of cellular communication such as registration
2. Describe how 802.11 supports mobile users
3. Describe practical uses of device-to-device communication, as well as multihop
4. Describe one type of mobile network such as ad hoc

NC-Emerging: Emerging Topics

KA Core:

1. Middleboxes (e.g., advances in usage of AI, intent-based networking, filtering, deep packet inspection, load balancing, NAT, CDN)
2. Network Virtualization (e.g., SDN, Data Center Networks)
3. Quantum Networking (e.g., Intro to the domain, teleportation, security, Quantum Internet)
4. Satellite, mmWave, Visible Light

Illustrative Learning Outcomes:

KA Core:

1. Describe the value of advances in middleboxes in networks.
2. Describe the importance of Software Defined Networks.
3. Describe some of the added value achieved by using Quantum Networking.

Professional Dispositions

- **Meticulous:** Students must be particular about the specifics of understanding and creating networking protocols.
- **Collaborative:** Students must work together to develop multiple components that interact together and to respond to failures and threats.
- **Proactive:** Students must be able to predict failures, threats, and how to deal with them while avoiding reactive modes of operation only.
- **Professional:** Students must comply with the needs of the community and their expectations from a networked environment, and the demands of regulatory bodies.